Finite-Time Tracking of Arbitrary Swarm Formations with Safety and Connectivity Certificates

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***Abstract*— Certifying finite arrival times and convergence rates is essential for coordinating multi-agent systems in time-critical missions. However, convergence to a desired formation does not, by itself, guarantee collision avoidance or communication connectivity during transients. Existing swarm models can achieve finite-time tracking of collective variables, but may not provide independent settling guarantees for agents at arbitrarily prescribed stations. This paper proposes a first-order control framework for tracking arbitrary, time-varying formation stations relative to a moving reference. A component-wise fractional-power feedback ensures exact finite-time station tracking with an explicit settling-time bound independent of the interaction graph. To address transient safety, a graph-dependent exponential bound on relative disagreement is derived that remains valid for time-varying and switching graphs. Combining this bound with the geometry of the prescribed stations yields sufficient conditions for pairwise collision avoidance and preservation of a connected spanning communication backbone. These conditions further provide a coupling-gain design rule based on the available geometric clearance throughout the maneuver. Numerical simulations illustrate finite-time tracking, transient formation deformation, and the effectiveness and conservatism of the proposed safety and connectivity certificates.**

***Index Terms*— Multi-agent systems, formation control, finite-time stability, collision avoidance, network connectivity.**

I. Introduction

Multi-agent systems enable coordinated tasks in which individual robots must move collectively while maintaining a prescribed spatial arrangement [1]. In many applications, the desired formation is asymmetric or time-varying, with each agent assigned a distinct station relative to a moving reference. Such tasks arise in coordinated transportation, autonomous navigation, and distributed sensing, where formation geometry may need to change dynamically while preserving the overall coordination structure.

Achieving the desired formation is not sufficient on its own. During a maneuver, agents may collide or neighbors may move beyond their communicating range even when the initial and desired formations are both feasible. These transient violations are particularly important in time-critical missions, where asymptotic convergence provides neither a finite settling time for each agent to a desired position, nor a guarantee that safety and communication constraints are maintained along the way. The challenge is therefore to achieve exact finite-time tracking of individually assigned stations while ensuring that the transient motion remains compatible with physical separation and communication requirements.

These objectives are closely coupled but need not be addressed through a single constrained control law. In particular, tracking errors affect relative agent positions, while their collective disagreement determines how far the actual inter-agent distances can deviate from those prescribed by the station trajectories. This observation motivates a complementary approach: retain a nominal finite-time tracking controller and use an analytical bound on relative tracking disagreement to certify collision avoidance and communication connectivity. Such a separation between convergence and transient certification can provide explicit design conditions without embedding safety constraints directly into the tracking feedback.

This paper develops a distributed first-order kinematic model for multi-agent systems with arbitrary prescribed C^1 station trajectories. The resulting velocity commands drive each agent to its assigned station in finite time, with a settling-time bound independent of the communication graph. A separate graph-dependent analysis establishes exponential decay of the disagreement component and translates this bound into sufficient geometric conditions for pairwise collision avoidance and persistence of a designer-selected connected spanning communication backbone. These conditions further yield an offline coupling-gain selection rule based on the planned station geometry and initial disagreement. The proposed model serves as a high-level kinematic guidance law, generating velocity setpoints for an underlying low-level controller or autopilot.

The main contributions are:

1) A distributed first-order kinematic model with an explicit, graph-independent finite-time settling bound for tracking arbitrary prescribed C^1 station trajectories.
2) A graph-dependent exponential disagreement bound valid for undirected time-varying or switching graphs, including disconnected modes.
3) Sufficient geometric conditions for pairwise collision avoidance and persistence of a connected spanning communication backbone throughout the transient.
4) An offline coupling-gain design inequality derived from the time-varying clearance margin, illustrated through numerical examples that distinguish formal certification from observed safe behavior.

II. Background and Problem Statement

Finite-time consensus and formation-control protocols establish convergence of networked agents under prescribed in-

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teraction structures [**?**], [**?**]. Collective-motion and kinematic swarm models, in contrast, typically characterize centroid motion, aggregation, or emergent formation behavior rather than independent tracking of arbitrary assigned stations [5]. Collision avoidance and connectivity preservation have also been studied using potential/energy functions, decentralized optimization, and control barrier functions [**?**], [**?**], [**?**], [4]. These approaches incorporate safety and connectivity constraints into the control design, whereas the present work derives a separate a priori certificate from bounds on tracking disagreement. The resulting conditions are sufficient and may be conservative: failure to satisfy the certificate does not necessarily imply a constraint violation. The analysis draws on Laplacian-based estimates for consensus over time-varying networks [**?**], [6].

Problem statement: Consider n agents with positions $x_i \in \mathbb{R}^d$ and prescribed C^1 station trajectories $x_{d,i}(t)$. Define the tracking errors $\delta_i = x_i - x_{d,i}$. Given a minimum pairwise separation d_{safe}, an operational communication range R_c, and a designer-selected connected spanning backbone $\mathcal{H}$, the objective is to formulate a distributed kinematic model that (i) generates velocity commands driving every δ_i exactly to zero by an explicit deadline independent of the communication graph; (ii) provides verifiable sufficient conditions for $|x_i - x_j| > d_{\text{safe}}$ for all pairs while every edge of $\mathcal{H}$ remains within R_c; and (iii) admits an offline coupling-gain selection rule based on the planned station geometry and initial disagreement.

III. Kinematic Model

Consider $n \geq 2$ agents with positions $x_i \in \mathbb{R}^d$. The desired station trajectory $x_{d,i}$ for each agent i is defined as

$$x_{d,i}(t) = r(t) + p_i(t), \qquad \delta_i(t) = x_i(t) - x_{d,i}(t), \quad (1)$$

where $r(t) \in \mathbb{R}^d$ is a reference trajectory and $p_i(t) \in \mathbb{R}^d$ is the desired offset of agent i. Unless stated otherwise, the station trajectories $x_{d,i}$ are continuously differentiable with locally bounded derivatives. This formulation allows the desired formation to translate and change shape over time.

The distributed kinematic guidance law is given by

$$\dot{x}_i = -(x_i - x_{d,i}) - \Phi_\beta(x_i - x_{d,i}) + \dot{x}_{d,i} - \sum_{j=1}^{n} w_{ij} M (x_i - x_j), \quad (2)$$

where $\dot{x}_i$ is the commanded agent velocity, Φ_β is a component-wise fractional-power feedback, w_{ij} are non-negative interaction weights, and $M \in \mathbb{R}^{d\times d}$ is a symmetric positive semi-definite coupling matrix. For $0 < \beta < 1$, the component-wise fractional-power feedback is defined by

$$[\Phi_\beta(z)]k = \text{sign}(z_k)|z_k|^\beta, \qquad k = 1, \ldots, d, \quad (3)$$

with $\Phi\beta(0) = 0$.

The interactions among agents are represented by an undirected weighted graph with symmetric weights $w_{ij} = w_{ji} \geq 0$ and $w_{ii} = 0$. Let $W = [w_{ij}]$ denote the weighted adjacency matrix and define the weighted graph Laplacian as

$$L_w = \text{diag}\left(\sum_{j=1}^{n} w_{ij}\right) - W. \quad (4)$$

For the range-dependent interaction model considered in this work, the weights are given by

$$w_{ij}(x,t) = \frac{a_{ij}(x,t)}{\|x_i - x_j\|^\nu + \varepsilon_w}, \qquad \nu \geq 1, \quad \varepsilon_w > 0, \quad (5)$$

where the communication indicator is determined strictly by the operational range,

$$a_{ij}(x) = \begin{cases} 1, & \|x_i - x_j\| \leq R_c, \\ 0, & \|x_i - x_j\| > R_c, \end{cases} \qquad a_{ij} = a_{ji}, \quad a_{ii} = 0. \quad (6)$$

Thus, the active communication graph is the range graph induced by the current agent positions; R_c is the operational communication radius. The positive constant ε_w prevents the weights from becoming singular when agents coincide. The designer also specifies a connected spanning backbone $\mathcal{H}$ (at minimum, a spanning tree) whose edges must remain active throughout the maneuver.

From (1), the relative tracking errors satisfy

$$\delta_i - \delta_j = (x_i - x_j) - (x_{d,i} - x_{d,j}), \quad (7)$$

so the coupling term can be computed using relative agent positions and desired relative station assignments.

Stacking the agent states and tracking errors as $x = [x_1^\top, \ldots, x_n^\top]^\top$ and $\delta = [\delta_1^\top, \ldots, \delta_n^\top]^\top$, respectively, the kinematic model (2) can be expressed in graph-based form as

$$\dot{\delta} = -\delta - \Phi_\beta(\delta) - (L_w(x,t) \otimes M)\delta, \quad (8)$$

where $\Phi_\beta(\delta)$ denotes the stacked component-wise fractional-power feedback.

Assumption 1. *Each agent knows its own state, reference and station trajectory (including $\dot{x}_{d,i}$), and exchanges its state and station information with currently active neighbors. Hence, each agent can compute δ_i and the relative errors $\delta_i - \delta_j$ for its active neighbors. The selected backbone $\mathcal{H}$ is specified by the designer and is a connected spanning subgraph of the planned communication network.*

IV. Finite-Time Station Tracking and Reference Convergence

Let $\delta_0 = \delta(t_0)$ and define the non-negative envelope

$$\mathcal{R}(t) = \left[\left(1 + \|\delta_0\|^{1-\beta}\right)e^{-(1-\beta)(t-t_0)} - 1\right]^{1/(1-\beta)}, \quad (9)$$

and finite-time arrival deadline be

$$T = t_0 + \frac{\ln(1 + \|\delta_0\|^{1-\beta})}{1-\beta}. \quad (10)$$

Theorem 1 (Finite-time assigned-station tracking)**.** *Under Assumption 1, with $M = M^\top \succeq 0$ and $0 < \beta < 1$, every solution of* (8) *satisfies*

$$\|\delta(t)\| \leq \mathcal{R}(t), \qquad t \geq t_0, \quad (11)$$

and $\delta(t) = 0$ for all $t \geq T$. Thus, $x_i(t) = x_{d,i}(t)$ for all agents and all $t \geq T$. The deadline is independent of the interaction graph, its connectivity, and M.

Proof. Consider $V = \frac{1}{2}\|\delta\|^2$. Symmetry of the weights and positive semi-definiteness of M give the edge-energy identity

$$\begin{aligned}\mathcal{D} &:= \delta^{\mathsf{T}}(L_w \otimes M)\delta \\ &= \frac{1}{2}\sum_{i,j} w_{ij}(\delta_i - \delta_j)^{\mathsf{T}} M(\delta_i - \delta_j) \geq 0. \end{aligned} \tag{12}$$

Since $1+\beta < 2$, norm monotonicity implies $\sum_{\ell=1}^{nd} |\delta_\ell|^{1+\beta} \geq \|\delta\|_2^{1+\beta}$. Consequently along (8),

$$\dot{V} = -\|\delta\|^2 - \sum_{i=1}^{n}\sum_{k=1}^{d} |\delta_{ik}|^{1+\beta} - \mathcal{D} \tag{13}$$

$$\leq -\|\delta\|^2 - \|\delta\|^{1+\beta} = -2V - (2V)^{(1+\beta)/2}, \tag{14}$$

Let $z = \|\delta\| = \sqrt{2V}$. Whenever $z > 0$, (14) implies $\dot{z} \leq -z - z^\beta$. With $y = z^{1-\beta}$,

$$\dot{y} \leq -(1-\beta)(y+1). \tag{15}$$

Integration yields $y(t) + 1 \leq (y(t_0) + 1)e^{-(1-\beta)(t-t_0)}$ until the tracking error reaches zero. This comparison bound establishes the bounds in (9) and (10), guaranteeing that the error reaches zero within a finite time. Since the origin is invariant under the error dynamics, the error remains zero after reaching it. Since V is non-increasing, the error remains bounded; the regularity and well-posed switching assumptions ensure the comparison holds across topology changes. □

A. Centroid and Formation Guarantees

Assume the station offsets are centered:

$$\sum_{i=1}^{n} p_i(t) = 0, \qquad \forall t \geq t_0. \tag{16}$$

Let $c(t) = \frac{1}{n}\sum_i x_i(t)$ be the swarm centroid. Since $x_{d,i} = r + p_i$, the mean desired station is r.

Corollary 1 (Reference tracking)**.** *Under the centering condition* (16)*,*

$$\|c(t) - r(t)\| \leq \frac{\mathcal{R}(t)}{\sqrt{n}}, \qquad t \geq t_0. \tag{17}$$

Consequently, $c(t) = r(t)$ for all $t \geq T$.

Proof. Since $x_{d,i} = r + p_i$ and $\sum_{i=1}^n p_i = 0$, the centroid error satisfies

$$c - r = \frac{1}{n}\sum_{i=1}^{n}(x_i - x_{d,i}) = \frac{1}{n}\sum_{i=1}^{n}\delta_i.$$

By the Cauchy–Schwarz inequality,

$$\|c - r\| \leq \frac{1}{\sqrt{n}}\left(\sum_{i=1}^{n}\|\delta_i\|^2\right)^{1/2} = \frac{\|\delta\|}{\sqrt{n}}.$$

Applying the bound $\|\delta(t)\| \leq \mathcal{R}(t)$ from Theorem 1 gives the stated inequality. Since $\mathcal{R}(t) = 0$ for all $t \geq T$, it follows that $c(t) = r(t)$ for all $t \geq T$. □

Remark 1. *The finite-time reference result is inherited from exact station tracking. If the offsets are not centered, the station theorem remains valid but the swarm centroid tracks the mean station $r + \frac{1}{n}\sum_i p_i$, not r.*

V. Disagreement-Based Safety and Connectivity Certificates

The complete station error contains both a common component and a disagreement component. Only disagreement changes relative positions. Define

$$\begin{aligned}&\Pi = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad \eta = (\Pi \otimes I_d)\delta, \\ &\mu = \lambda_{\min}(M) \geq 0.\end{aligned} \tag{18}$$

Thus $\eta_i = \delta_i - \bar{\delta}$ and $\delta_i - \delta_j = \eta_i - \eta_j$.

Proposition 1 (Graph-dependent disagreement contraction)**.** *For the undirected weighted graph in* (5)*, every solution satisfies*

$$\|\eta(t)\| \leq \|\eta(t_0)\| \exp\left[-\int_{t_0}^{t}(1 + \mu\lambda_2(L_w(s)))\, ds\right] \tag{19}$$

Here $\lambda_2(L_w)$ is the algebraic connectivity, with $\lambda_2 = 0$ for a disconnected graph and $\mu = \lambda_{\min}(M)$.The bound remains valid through admissible topology switches.

Proof. Since $\Pi L_w = L_w$ and $\Pi\eta = \eta$, the disagreement dynamics followed from (8) are

$$\dot{\eta} = -\eta - (\Pi \otimes I_d)\Phi_\beta(\delta) - (L_w \otimes M)\eta. \tag{20}$$

For $W_\eta = \frac{1}{2}\|\eta\|^2$, use $\eta^{\mathsf{T}}(\Pi \otimes I_d) = \eta^{\mathsf{T}}$ to obtain

$$\dot{W}_\eta = -\|\eta\|^2 - \eta^{\mathsf{T}}\Phi_\beta(\delta) - \eta^{\mathsf{T}}(L_w \otimes M)\eta. \tag{21}$$

The non-linear term is non-negative because component-wise monotonicity of Φ_β implies

$$\eta^{\mathsf{T}}\Phi_\beta(\delta) = \frac{1}{2n}\sum_{i,j}(\delta_i - \delta_j)^\top\big(\Phi_\beta(\delta_i) - \Phi_\beta(\delta_j)\big) \geq 0 \tag{22}$$

Since $M \succeq 0$ and $\mu = \lambda_{\min}(M)$, the Rayleigh quotient bound gives

$$\eta^\top(L_w \otimes M)\eta \geq \mu\lambda_2(L_w)\|\eta\|^2$$

Discarding the non-positive term,

$$\dot{W}_\eta \leq -2\big[1 + \mu\lambda_2(L_w)\big]W_\eta.$$

Applying the comparison lemma and integrating yields (19). □

A. Pairwise Distance Bounds

Let $d_{ij}^q(t) = \|x_{d,i}(t) - x_{d,j}(t)\|$ and $d_{ij}^x(t) = \|x_i(t) - x_j(t)\|$.

Lemma 1 (Relative-distance envelope)**.** *For every pair i, j,*

$$d_{ij}^q(t) - \sqrt{2}\,\|\eta(t)\| \le d_{ij}^x(t) \le d_{ij}^q(t) + \sqrt{2}\,\|\eta(t)\|\,. \quad (23)$$

Proof. Since $x_i - x_j = (x_{d,i} - x_{d,j}) + (\eta_i - \eta_j)$, the triangle inequality gives

$$\|x_i - x_j\| \le \|x_{d,i} - x_{d,j}\| + \|\eta_i - \eta_j\|,$$

while the reverse-triangle inequality yields

$$\|x_i - x_j\| \ge \|x_{d,i} - x_{d,j}\| - \|\eta_i - \eta_j\|.$$

Moreover,

$$\|\eta_i - \eta_j\| \le \|\eta_i\| + \|\eta_j\| \le \sqrt{2(\|\eta_i\|^2 + \|\eta_j\|^2)} \le \sqrt{2}\,\|\eta\|\,.$$

Hence, the two inequalities above establishes the claimed bounds. □

A connected spanning backbone $\mathcal{H} = (\mathcal{V}, \mathcal{E}_\mathcal{H})$ is specified by the designer to preserve communication links essential for coordination during formation maneuvers. Let $L_\mathcal{H}$ denote its unweighted Laplacian. The station-edge length envelope and the minimum pairwise station distance are defined as

$$\begin{aligned} \ell_\mathcal{H}(t) &= \max_{(i,j)\in\mathcal{E}_\mathcal{H}} \|x_{d,i}(t) - x_{d,j}(t)\|\,, \\ d_q(t) &= \min_{i<j} \|x_{d,i}(t) - x_{d,j}(t)\|\,. \end{aligned} \quad (24)$$

Let $\lambda_2(L_\mathcal{H}) > 0$ and let $\underline{w} > 0$ be a lower bound on every active backbone-edge weight. For the range-dependent weights in (5), one may choose $\underline{w} = (R_c^\nu + \varepsilon_w)^{-1}$. Define

$$\begin{aligned} \gamma_\mathcal{H} &= 1 + \mu\underline{w}\lambda_2(L_\mathcal{H}), \\ \rho_0 &= \sqrt{2}\,\|\eta(t_0)\|\,. \end{aligned} \quad (25)$$

Since $\eta = (\Pi \otimes I_d)\delta$ and Π is an orthogonal projection, it cannot increase the Euclidean norm; hence $\|\eta(t)\| \le \|\delta(t)\|$. By Theorem 1, this gives $\sqrt{2}\|\eta(t)\| \le \sqrt{2}\,\mathcal{R}(t)$. Moreover, Proposition 1 gives exponential contraction of the disagreement. While all backbone edges remain active, $L_w \succeq \underline{w}L_\mathcal{H}$, so the contraction rate is at least $\gamma_\mathcal{H}$. Consequently,

$$\rho_\mathcal{H}(t) = \min\left\{\sqrt{2}\,\mathcal{R}(t),\, \rho_0 e^{-\gamma_\mathcal{H}(t-t_0)}\right\}. \quad (26)$$

This is an a priori envelope on the maximum pairwise disagreement for every pair of agents, provided the backbone remains available.

Theorem 2 (Joint collision avoidance and connectivity preservation)**.** *If the planned stations satisfy, for all $t \ge t_0$,*

$$d_q(t) > d_{\text{safe}} + \rho_\mathcal{H}(t), \quad (27)$$

$$\ell_\mathcal{H}(t) < R_c - \rho_\mathcal{H}(t), \quad (28)$$

then every pair of agents remains strictly separated by more than d_{safe}, every backbone edge remains within range R_c, and the communication graph remains connected. The station-tracking deadline T in (10) *is unchanged.*

Proof. At t_0, (26) gives $\rho_\mathcal{H}(t_0) = \rho_0$, since $\|\eta(t_0)\| \le \|\delta(t_0)\|$. By Lemma 1 and (27), all actual pairwise distances exceed d_{safe} initially. The same lemma and (28) put all backbone links strictly inside R_c initially.

Suppose a backbone edge first reaches R_c at time $t_e > t_0$. Until t_e, all backbone edges remain active, so the lower bound $L_w \succeq \underline{w}L_\mathcal{H}$ and Proposition 1 imply (26). By continuity, the bound also holds at t_e. For every backbone edge, Lemma 1 then gives $\|x_i(t_e) - x_j(t_e)\| \le \ell_\mathcal{H}(t_e) + \rho_\mathcal{H}(t_e) < R_c$, a contradiction. Thus the backbone persists for all time. Finally, the lower half of Lemma 1 and (27) imply $d_{ij}^x(t) > d_{\text{safe}}$ for every pair and all times. The station-tracking theorem is unaffected by the graph argument. □

B. Coupling-Gain Design Rule

Define the available geometric margin

$$s(t) = \min\{d_q(t) - d_{\text{safe}},\, R_c - \ell_\mathcal{H}(t)\}. \quad (29)$$

The margin depends only on the prescribed station trajectories, the selected safety threshold, the operational communication range, and the chosen backbone. It can therefore be computed offline, before execution, without live agent-state measurements. Geometric feasibility requires $s(t) > 0$ throughout the interval of interest. Suppose $s(t) > 0$ on $[t_0, T]$ and $0 < \rho_0 < s(t_0)$. The latter is an initial-margin condition: if it fails, increasing the coupling gain cannot make this particular certificate applicable. Define

$$\gamma_{\text{req}} = \sup_{t_0<t<T} \frac{\ln\big(\rho_0/s(t)\big)}{t - t_0}. \quad (30)$$

Corollary 2 (Sufficient coupling selection)**.** *A sufficient condition for Theorem 2 is*

$$1 + \lambda_{\min}(M)\,\underline{w}\,\lambda_2(L_\mathcal{H}) > \gamma_{\text{req}}. \quad (31)$$

For $M = mI_d$, if $\gamma_{\text{req}} > 1$, it is sufficient to choose

$$m > \frac{\gamma_{\text{req}} - 1}{\underline{w}\lambda_2(L_\mathcal{H})}. \quad (32)$$

Thus, once the station plan and physical limits (d_{safe}, R_c) are specified, the margin and required rate can be computed offline and used to select a gain satisfying the certificate.

Proof. By Theorem 2, it suffices to ensure that $\rho_\mathcal{H}(t) < s(t)$ throughout the maneuver. By the definition of γ_{req} in (30), $\gamma_\mathcal{H} > \gamma_{\text{req}}$ ensures that

$$\rho_0 e^{-\gamma_\mathcal{H}(t-t_0)} < s(t), \qquad t \in (t_0, T).$$

Since $\rho_\mathcal{H}(t) \le \rho_0 e^{-\gamma_\mathcal{H}(t-t_0)}$ by (26), the sufficient conditions hold on (t_0, T). The strict initial condition handles $t = t_0$, while at $t = T$, $\rho_\mathcal{H}(T) = 0 < s(T)$. Thus, the certificate holds throughout the maneuver. In particular, its minimum margin

$$\Delta_{\min} := \min_{t\in[t_0,T]} [s(t) - \rho_\mathcal{H}(t)] \quad (33)$$

is strictly positive, quantifying the available clearance of the certificate. □

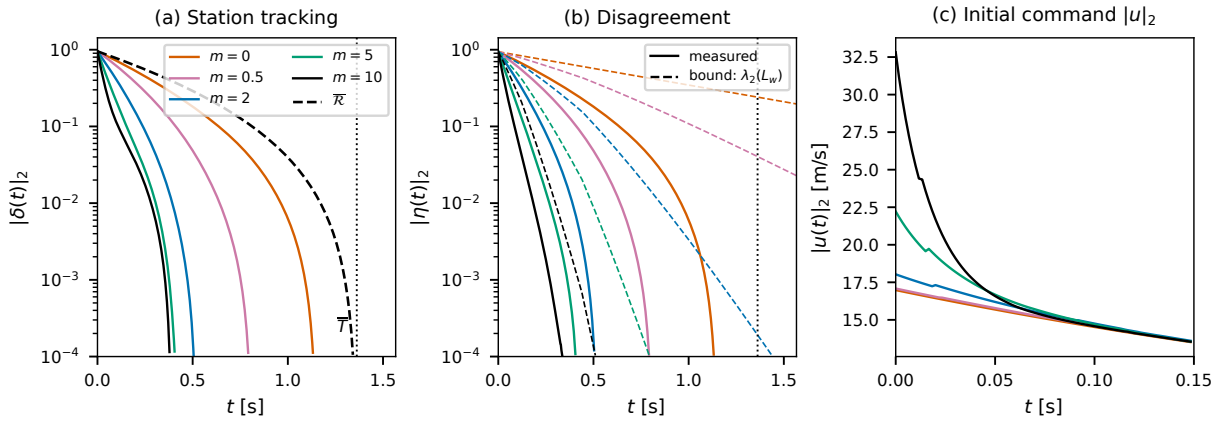


Fig. 1: Finite-time tracking, disagreement decay, and control effort across coupling gains m ($\kappa = 2.0$): (a) $\|\delta(t)\|_2$ and theoretical bound $\overline{\mathcal{R}}(t)$ with deadline T; (b) $\|\eta(t)\|_2$ and analytical envelope bounds; (c) $\|u(t)\|_2$ over the initial 0.15 s.

TABLE I: Benchmark simulation parameters.

Param.	Value	Description
n	10	Swarm size in $\mathbb{R}^2$
β	0.5	Fractional-power velocity exponent
d_{safe}	1.0 m	Minimum pairwise collision threshold
R_c	6.0 m	Maximum communication radius
ε_w	0.05	Proximity weight regularizer
ν	1	Distance exponent in interaction weights
$\underline{w}$	0.1653 m^{-1}	Uniform lower bound $1/(R_c + \varepsilon_w)$
$r(t)$	$[t,\ 2\sin(0.5t)]^{\mathsf{T}}$	Reference centroid trajectory
ω	$\pi/4$ rad/s	Star template angular rotation rate
M	mI_2	Cooperative coupling matrix ($m \geq 0$)

Remark 2. *If $\gamma_{\text{req}} < 1$, the baseline term satisfies the rate condition with $M = 0$. The gain condition is sufficient, not optimal: its failure does not imply a constraint violation, and increasing coupling cannot compensate for a failed initial-margin test $\rho_0 < s(t_0)$. Moreover, large gains may require velocity commands beyond vehicle limits, which must be considered in practical gain selection.*

VI. Numerical Validation

The theoretical results are evaluated through simulations of $n = 10$ agents in $\mathbb{R}^2$ using the benchmark parameters in Table I. The complete simulation framework and experiments are available at `https://github.com/yash-purswani/finite-time-swarm-control`.

Three scenarios are considered: (i) finite-time tracking and coupling-gain trade-offs, (ii) aggressive star compression and expansion to test transient safety and connectivity, and (iii) narrow-corridor traversal to demonstrate shape morphing. Unless stated otherwise, initial perturbations satisfy $\rho_0 < s(0)$. The star formation rotates at rate ω and scales as $a(t) = a_\infty + (a_0 - a_\infty)e^{-\kappa t}$. Connectivity is assessed using a 15-edge spanning backbone with $\lambda_2(L_{\mathcal{H}}) \approx 0.9316$, yielding the certified disagreement decay rate $\gamma_{\mathcal{H}} = 1 + 0.1540m$.

A. Finite-Time Tracking and Coupling-Gain Trade-offs

The influence of coupling gain m is evaluated during rotating star compression ($\kappa = 2.0$, $a_0 = 1.70$ to $a_\infty = 0.72$). As shown in Figure 1 and Table II, the tracking error satisfies the common theoretical deadline for all tested gains. The reported $t_{10^{-6}}$ is the first time at which $\|\delta(t)\|_2 \leq 10^{-6}$; it is a numerical tolerance metric, not verification of exact equality. Increasing m accelerates disagreement decay, while the theoretical deadline remains unchanged.

TABLE II: Settling times and control metrics across coupling gains m under rotating star formation ($\kappa = 2.0$).

m	$t_{10^{-6}}$ (s)	T (s)	Peak $\|u\|_2$	J_u
0.0	1.1517	1.3621	16.9828	429.44
0.5	0.8117	1.3621	17.0860	429.92
2.0	0.5250	1.3621	18.0322	431.55
5.0	0.4467	1.3621	22.2186	434.95
10.0	0.3933	1.3621	32.8156	440.74

B. Transient Safety and Connectivity

Rapid star compression and expansion are evaluated under identical maneuver rate $\kappa = 12.0$ and coupling gain $m = 60.0$ ($m = 1.0$ for the unsafe comparisons), distinguished solely by their scaling targets: compression contracts the swarm from $a_0 = 1.70$ to $a_\infty = 0.72$, whereas expansion expands it from $a_0 = 1.70$ to $a_\infty = 3.02$. The geometric margins, disagreement envelopes, and observed pairwise distances are shown in Figure 2; quantitative outcomes and minimum certificate margins are reported in Table III. Under both maneuvers, the coupling gain $m = 60.0$ yields $\gamma_{\mathcal{H}} = 10.2388$, strictly exceeding the required certification rates ($\gamma_{\text{req}} = 9.9879$ for compression, $\gamma_{\text{req}} = 6.3408$ for expansion). Consequently, both maneuvers satisfy the offline certificate ($\Delta_{\min} = 0.0084$ m > 0 and $\Delta_{\min} = 0.1113$ m > 0, respectively), guaranteeing that pairwise distances remain strictly within bounds: collision-free ($d_{x,\min} = 1.0326$ m $> d_{\text{safe}}$) and connected ($\ell_{x,\max} = 5.8880$ m $< R_c$). In sharp contrast, the weakly coupled cases ($m = 1.0$) suffer genuine physical constraint breaches during the transient: a collision penetration under compression ($d_{x,\min} = 0.8923$ m $<$ 1.0 m) and a transient communication-range violation under

expansion ($\ell_{x,\max} = 6.2417$ m > 6.0 m).

C. Arbitrary Formation Morphing and Constrained Corridor Traversal

The final scenario combines time-varying station tracking, geometric feasibility, collision avoidance, and backbone connectivity in a corridor maneuver. The swarm traverses a corridor ($x \in [12, 30]$ m, half-width $y_{\text{wall}} = 2$ m) by smoothly blending, using C^2 quintic polynomials, from a circular ring into a slender chevron and back to a ring after exiting (Figure 3). Connectivity is assessed on a designer-selected 10-cycle backbone $\mathcal{H}$. For this planned maneuver, the initial-margin and geometric-feasibility conditions are satisfied, and the certificate remains satisfied over the maneuver. The agents reach their stations by the finite-time deadline ($T \approx 1.14$ s); the observed minimum separation is at least 1.32 m and the maximum backbone-edge length is at most 2.22 m. This example therefore illustrates the combined use of the tracking controller and the safety/connectivity certificate, rather than only an empirical corridor traversal.

VII. Conclusion

This paper presented a distributed first-order controller that tracks arbitrary C^1 station trajectories in finite time with a graph-independent deadline. A separate disagreement estimate bounds deviations of actual pairwise distances from the prescribed station geometry. For a designer-selected connected spanning backbone, this bound yields sufficient conditions for collision avoidance and communication-range preservation, together with an offline coupling-gain rule based on the planned geometric margin and initial disagreement. The compression, expansion, and corridor scenarios illustrate both the usefulness and the conditional nature of the certificate: satisfying it guarantees the stated constraints under the assumptions, whereas failing it is inconclusive and does not by itself imply a violation. The analysis is based on ideal first-order kinematics and does not account for actuator saturation or other physical input constraints.

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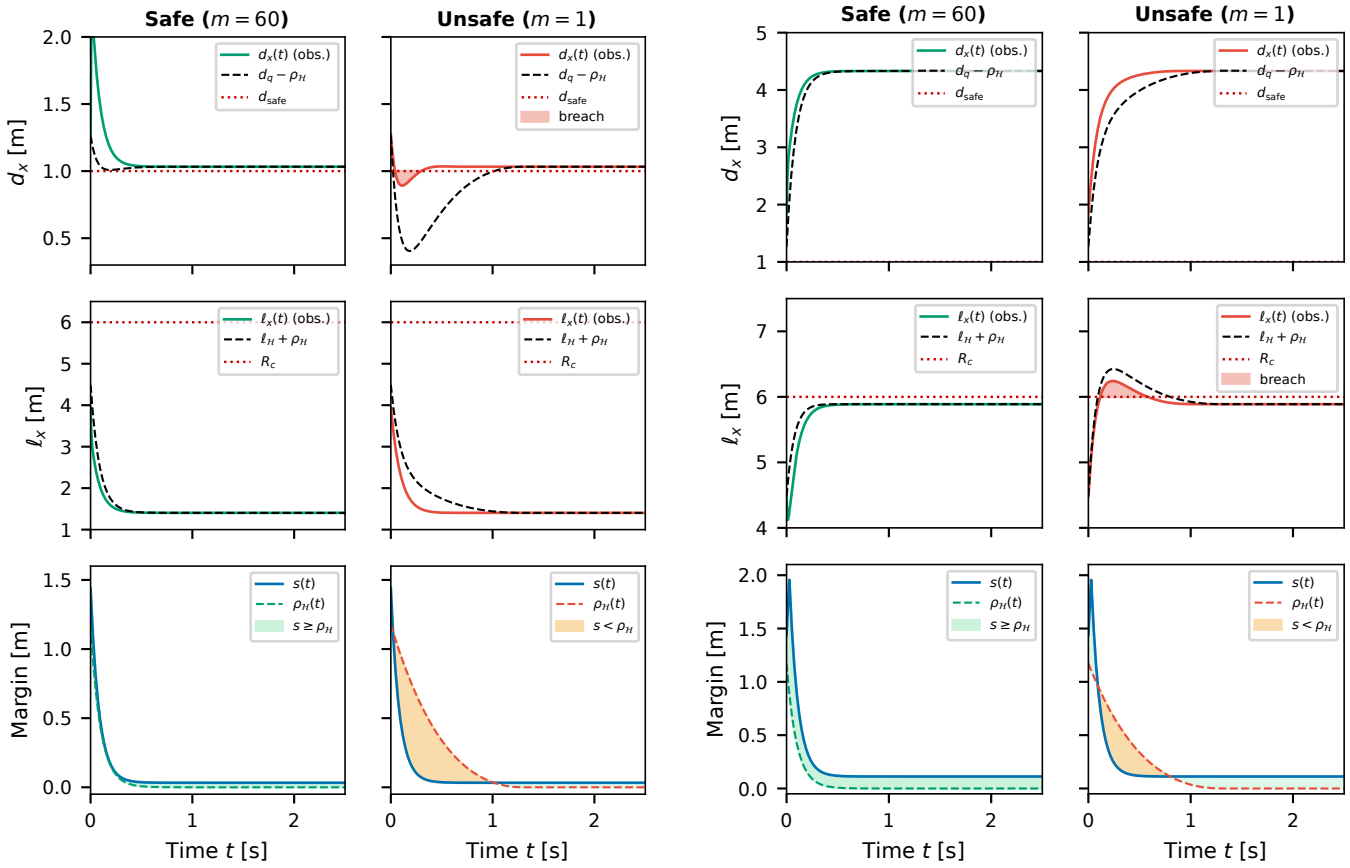
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(a) Formation compression ($a_0 = 1.70 \to a_\infty = 0.72$).

(b) Formation expansion ($a_0 = 1.70 \to a_\infty = 3.02$).

Fig. 2: Transient safety and connectivity verification across aggressive compression and expansion maneuvers ($\kappa = 12.0$, $t \in [0, 2.5]$ s). In both panels, columns show the minimum pairwise separation $d_x(t)$ (left), maximum pairwise distance $\ell_x(t)$ (middle), and geometric margin $s(t)$ versus disagreement tube $\rho_H(t)$ (right). **Left:** Star compression ($a_0 = 1.70 \to a_\infty = 0.72$); safe case ($m = 60.0$, certified collision-free), unsafe case ($m = 1.0$, collision breach observed). **Right:** Star expansion ($a_0 = 1.70 \to a_\infty = 3.02$); safe case ($m = 60.0$, certified connected), unsafe case ($m = 1.0$, transient connectivity violation observed).

TABLE III: Certificate and observed transient-geometry outcomes for aggressive star maneuvers ($\kappa = 12.0$, $t \in [0, 2.5]$ s).

Maneuver	Case	m	κ	$\gamma_{\mathcal{H}}$	γ_{req}	$t_{10^{-6}}$ (s)	$d_{x,\min}$ (m)	$\ell_{x,\max}$ (m)	$\Delta_{\min}$ (m)	Observed outcome
Compression	Safe	60.0	12.0	10.2388	9.9879	0.1767	1.0326	3.9437	0.0084	Safe and connected
	Unsafe	1.0	12.0	1.1540	9.9879	0.5450	0.8923	3.9437	≤ 0	Collision observed
Expansion	Safe	60.0	12.0	10.2388	6.3408	0.2417	1.8351	5.8880	0.1113	Safe and connected
	Unsafe	1.0	12.0	1.1540	6.3408	1.0433	1.8351	6.2417	≤ 0	Range violation observed

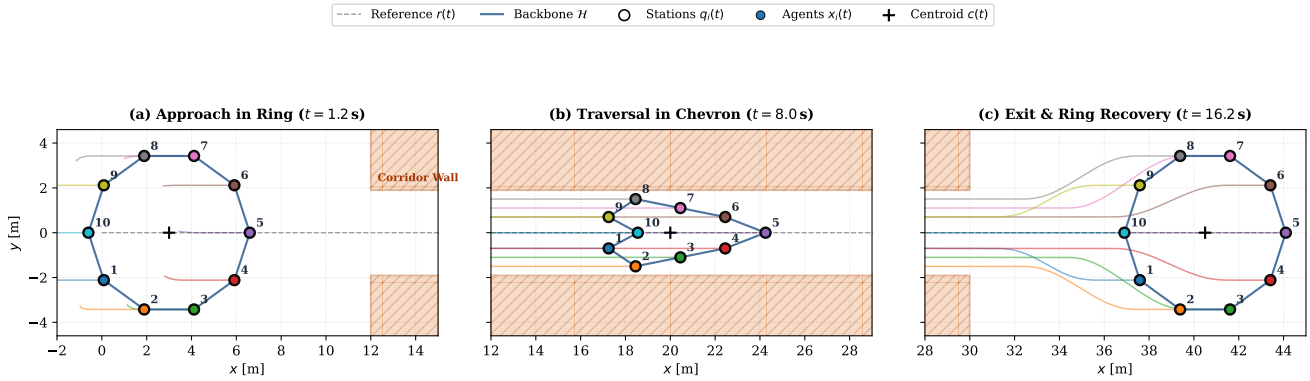


Fig. 3: Arbitrary formation morphing and constrained corridor traversal ($n = 10$, $m = 10.0$). Snapshots depict: (a) approach and finite-time convergence in ring geometry ($t = 1.2$ s), (b) morphing into a slender chevron to navigate the narrow corridor ($t = 8.0$ s), and (c) exit and ring recovery ($t = 16.2$ s). Target stations $x_{d,i}$ (circles), agent positions x_i (colored dots), continuous trajectories, reference centroid $r(t)$ (dashed line), and the 10-cycle backbone $\mathcal{H}$ (solid blue edges) remain strictly collision-free ($d_{x,\min} \geq 1.32$ m) and connected ($\ell_{x,\max} \leq 2.22$ m).